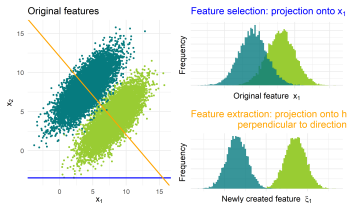
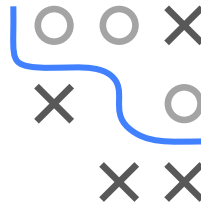


Feature Selection: Introduction



- Too many features can be harmful in prediction
- Selection vs. extraction
- Types of selection methods

INTRODUCTION

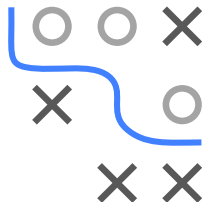
Feature selection:

Finding a well-performing, hopefully small set of features for a task.

Feature selection is critical for

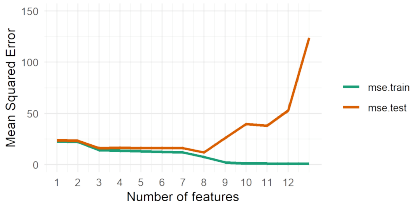
- reducing noise and overfitting
- improving performance/generalization
- enhancing interpretability by identifying most informative features

Features can be selected based on domain knowledge, or data-driven algorithmic approaches. We focus on the latter here.



MOTIVATION

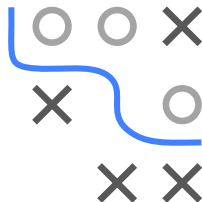
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SIZE OF DATASETS

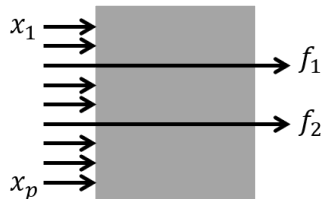
Many new forms of technical measurements and connected data leads to availability of extremely high-dimensional data sets.

- **Classical setting:** Up to around 10^2 features, feature selection might be relevant, but benefits often negligible.
- **Datasets of medium to high dimensionality:** At around 10^2 to 10^3 features, classical approaches can still work well, while principled feature selection helps in many cases.
- **High-dimensional data:** 10^3 to 10^9 or more features. Examples: micro-array / gene expression data and text categorization (bag-of-words features). If we also have few observations, scenario is called $p \gg n$.



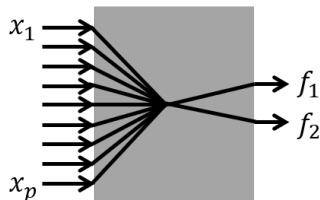
FEATURE SELECTION VS. EXTRACTION

Feature selection

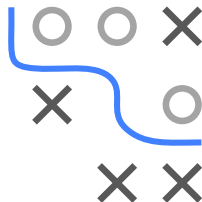


- Creates a subset of original features $\mathbf{x}$ by selecting $\tilde{p} < p$ features $\mathbf{f}$.
- Retains information on selected individual features.

Feature extraction



- Maps p features in $\mathbf{x}$ to $\tilde{p}$ extracted features $\mathbf{f}$.
- Info on individual features can be lost through (non-)linear combination.



TYPES OF FEATURE SELECTION METHODS

In rest of the chapter, we introduce different types of methods for FS:

- Filters: evaluate relevance of features using statistical properties such as correlation with target variable
- Wrappers: use a model to evaluate subsets of features
- Embedded methods: integrate FS directly into specific model - we look at them in their dedicated chapters (e.g., CART, L_0 , L_1)

Example: embedded method (Lasso) regularizing model params with L_1 penalty enables "automatic" feature selection:

$$\mathcal{R}_{\text{reg}}(\theta) = \mathcal{R}_{\text{emp}}(\theta) + \lambda \|\theta\|_1 = \sum_{i=1}^n (y^{(i)} - \theta^\top \mathbf{x}^{(i)})^2 + \lambda \sum_{j=1}^p |\theta_j|$$

